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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/972,577	12/04/2020	Jae Jun KO	4016-1056	2154

20311 7590 08/31/2021

LUCAS & MERCANTI, LLP
30 BROAD STREET
21st FLOOR
NEW YORK, NY 10004

EXAMINER
BOOSALIS, FANI POLYZOS

ART UNIT	PAPER NUMBER
2884	

NOTIFICATION DATE	DELIVERY MODE
08/31/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

info@lmiplaw.com

NON-FINAL OFFICE ACTION
Response Due: November 30, 2021
1st Extension Due: December 31, 2021
2nd Extension Due: January 31, 2022
3rd Extension and Final Due Date: February 28, 2022

Office Action Summary	Application No. 16/972,577	Applicant(s) KO et al.	
	Examiner FANI P BOOSALIS	Art Unit 2884	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on December 4, 2020.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-4,6-7,9-10,17 and 20 is/are rejected.

8) ☑ Claim(s) 5,8,11-16 and 18-19 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 12/4/2020 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 12/4/2020.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Claim Rejections - 35 USC § 102

2. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

3. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

1. Claim(s) 1, 6-7, 9, 10, 17, 20 is/are rejected under 35 U.S.C. 102(a)(2) as being anticipated by Kwon et al (US 2020/0278457 A1).

Regarding claim 1, Kwon et al discloses a radon sensor (100) using a polyhedral-shaped ionization chamber (See Fig. 1), the radon sensor device comprising: an ionization chamber (110) having an open side and inner sides (paragraph [0042]) surrounded by a first conductor and generating an electrical field therein by applying bias power to the first conductor (paragraphs [0039], [0041]); a cover (124) having a first side covered with a second conductor and closing the open side of the ionization chamber such that the first conductor disposed on the inner side of the ionization chamber disposed on the inner sides of the ionization chamber and the second conductor are electrically connected (i.e. cover member made of a conductive material and portions made of conductive material may be in contact

with the ionization chamber (110), at least one surface, that is, any one of upper and lower end surfaces of the cover member may be made of plastic material and then plated with a conductive material so as to have conductivity and an edge of the plated surface may be in contact with the stepped portion (114) of the ionization chamber) (See Figs. 2-3, 5 and paragraph [0047]); a probe unit (130) disposed in the ionization chamber (110) and absorbing ion charges produced when alpha (α) decay occurs in the ionization chamber (See Fig. 3 and paragraphs [0039], [0058]); and a measurement circuit (100) detecting an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit into a predetermined magnitude (paragraphs [0011]-[0012], [0063]).

Regarding claim 6, Kwon et al discloses wherein the cover is a printed circuit board (PCB) having a predetermined thickness and the entire of a first side of the printed circuit board is covered with the second conductor (See Fig. 6 and paragraph [0067]).

Regarding claim 7, Kwon et al discloses wherein a plurality of holes (112) is formed through the cover so that external air can flow into the ionization chamber or air can flow outside (See Figs. 1-2, paragraph [0042]).

Regarding claim 9, Kwon et al discloses wherein the probe unit detects ion charges produced by a collision of air and alpha particles produced when alpha (α) decay occurs in the ionization chamber (See Figs. 2-4 and paragraph [0058]).

Regarding claim 10, Kwon et al discloses wherein the probe unit (130) is disposed at a center in the ionization chamber and is formed in a T-shape (See Fig. 3).

Regarding claim 17, Kwon et al discloses wherein the measurement circuit is disposed on a second side of the cover (124) and is mounted to be electrically connected with the second conductor (160) disposed on the first side of the cover through a bias voltage outputter for outputting the bias power (paragraph [0051]).

Regarding claim 20, Kwon et al discloses an electromagnetic wave-protective cap (150) installed on a second side of the cover (124) to surround the entire measurement circuit and is made of a conductive material to prevent signal disturbance of the measurement circuit due to external electromagnetic wave noise, wherein the electromagnetic wave-protective cap is installed to be electrically connected with a ground of the measurement circuit (See Fig. 2 and paragraph [0048]).

Claim Rejections - 35 USC § 103

2. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

3. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

4. Claim 2 is/are rejected under 35 U.S.C. 103 as being unpatentable over Kwon et al (US 2020/0278457 A1) in view of Wedding et al (US 7,791,037 B1).

Regarding claim 2, Kwon et al discloses all of the limitations of parent claim 1, as described above, however, Kwon et al is silent with regards to the chamber formed in a hexahedral shape. Wedding et al discloses a radiation detection device (800), comprising: a plasma tube (chamber) of any suitable geometric shape, hexagon, pentagon, square, and so forth using an insulating material (col. 10, lines 31-44). Thus, it would have been obvious to modify Kwon et al to comprise a chamber in a hexahedral shape, as taught supra by Wedding et al, so as to enable good mountability and high measurement efficiency without possibility of electric hock or leakage of electricity.

5. Claim 3 is/are rejected under 35 U.S.C. 103 as being unpatentable over Kwon et al (US 2020/0278457 A1) in view of Barstad et al (US 2,842,674

Regarding claim 3, Kwon et al discloses all of the limitations of parent claim 1, as described above, however, Kwon et al is silent with regards to coating conductive paint as claimed. Barstad et al discloses a radioactive radiation indicator, comprising: a chamber (4) wherein a first conductor disposed on the inner sides of the ionization chamber is formed by coating conductive paint (graphite paint) (5) with a predetermined thickness on the inner sides of the ionization chamber (See Fig. 1 and col. 4, lines 15-25). Thus, it would have been obvious to modify Kwon et al to use a conductive coating, as taught supra by Barstad et al, so as to prevent signal disturbance.

6. Claim 4 is/are rejected under 35 U.S.C. 103 as being unpatentable over Kwon et al (US 2020/0278457 A1) in view of Esin et al (US 8,415,638 B2).

Regarding claim 4, Kwon et al discloses all of the limitations of parent claim 1, as described above, however, Kwon et al is silent with regards to the DC voltage range, as claimed. Esin et al discloses a method and detecting high-energy radiation using low voltage optimized ion chamber, comprising: bias power applied to a first conductor disposed on the inner sides of the ionization chamber (13) is a DC voltage of 50V to 300V (i.e. 120-1500V DC) (col. 4, lines 10-14 and col. 6, lines 51-58). Thus, it would have been obvious to modify Kwon et al to enable voltage in the range of 50V to 300V, as claimed supra by Esin et al so as to enable a reliable and accurate measurement of high energy radiation.

Allowable Subject Matter

7. Claims 5, 8, 11-16, 18-19 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Regarding claim 5, the prior art fails to disclose or reasonably suggest wherein the first conductor disposed on the inner sides of the ionization chamber and the second conductor disposed on the first side of the cover are electrically connected through joints between the cover and the ionization chamber, as claimed in combination with the rest of the claim limitations.

Regarding claim 8, the prior art fails to disclose or reasonably suggest wherein when a volume of the ionization chamber is 100cc, a diameter of the holes formed through the cover is 0.2mm to 0.3mm and the number thereof is 6 to 10, as claimed in combination with the rest of the claim limitations.

Regarding claim 11, the prior art fails to disclose or reasonably suggest wherein a center of the second probe is perpendicularly coupled to the second end of the first probe and is disposed at the center in the ionization chamber such that distances between the inner sides of the ionization chamber and a surrounding portion of the second probe are the same, as claimed in combination with the rest of the claim limitations.

Regarding claims 12-16, the prior art fails to disclose or reasonably suggest wherein the probe unit is disposed at a center in the ionization chamber formed in an outdoor TV antenna shape; wherein the second probes are formed in a rod shape or a straight shape with a predetermined length, are coupled at centers to the upper head of the first probe, are longitudinally spaced a predetermined distance from each other, and are coupled perpendicular to or at a predetermined angle with respect to a longitudinal direction of the upper head of the first probe to have a predetermined pattern; wherein the probe unit is disposed in the ionization chamber such that distances between the inner sides of the ionization chamber and a surrounding portion of the second probe are the same, as claimed in combination with the rest of the claim limitations.

Regarding claims 18-19, the prior art fails to disclose or reasonably suggest wherein a noise detector connected between the guard ring and the ground, sends a DC-type leakage current generated between the lonization chamber and the probe to the ground, and detects and outputs AC-type noise;

first and second pre-amplifiers respectively connected to another end of the probe and an output end of the noise detector, and amplifying small electrical signals input from the probe and the noise detector into a predetermined magnitude; and a differential amplifier having a non-inverting terminal (+) and an inverting terminal (-) respectively connected to output ends of the first and second pre-amplifiers, and offsetting noise signals and outputting an alpha particle detection signal by amplifying a voltage difference of electrical signals pre-amplified through the first and second pre-amplifiers, as claimed in combination with the rest of the claim limitations.

Conclusion

4. Any inquiry concerning this communication or earlier communications from the examiner should be directed to FANI POLYZOS BOOSALIS whose telephone number is (571)272-2447. The examiner can normally be reached on 7:30-3:30 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, David Porta can be reached on 571-272-2444. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

5. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see https://ppair-my.uspto.gov/pair/PrivatePair. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a

USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/DAVID P PORTA/
Supervisory Patent Examiner, Art Unit 2884

/FB/

Notice of References Cited	Application/Control No. 16/972,577	Applicant(s)/Patent Under Reexamination KO et al.	
	Examiner FANI P BOOSALIS	Art Unit 2884	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20200278457-A1	09-2020	KWON; Ji Sung	G01T1/178	1/1
*	B	US-4755682-A	07-1988	Burgess; Edward T.	G01T1/185	250/374
*	C	US-7791037-B1	09-2010	Wedding; Carol Ann	G01T1/185	250/374
*	D	US-2842674-A	07-1958	BARSTAD GOTFRED E B	G01T1/185	250/378
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FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
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	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (01-10)
Approved for use through 07/31/2012. OMB 0651-0031
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT

(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	KO, Jae Jun
Art Unit	TBA
Examiner Name	TBA
Attorney Docket Number	4016-1056

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Examiner Initial*	Cite No	Patent Number	Kind Code[1]	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
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Examiner Initial*	Cite No	Foreign Document Number[3]	Country Code[2] i	Kind Code[4]	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T[5]
	1	101730887	KR	B1	2017-04-28	FTLAB CO., LTD.	English Abstract	
	2	101730891	KR	B1	2017-04-28	FTLAB CO., LTD.	English Abstract	
	3	10-1935880	KR	B1	2019-01-17	ISQUARE, INC.	English Abstract	

 /FANI POLYZOS BOOSALIS/ 08/25/2021

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /F.P.B./

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Examiner Name	TBA
Attorney Docket Number	4016-1056

	4	10-2015-0093987	KR	A	2015-08-19	LISOTEK CO., LTD. et al.	English Abstract	
	5	10-2017-0023599	KR	A	2017-03-06	FTLAB CO., LTD.	English Abstract	
	6	2015-156463	JP	A	2005-06-16	MITSUBISHI HEAVY IND., LTD.	English Abstract	
	7	10-2018-0129480	KR	A	2018-12-05	LEE, Hoon	English Abstract	

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NON-PATENT LITERATURE DOCUMENTS Remove

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T[5]
	1	International Search Report and Written Opinion for International application No. PCT/KR2020/000837, April 24, 2020, ISA/KR	

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EXAMINER SIGNATURE

Examiner Signature	/FANI POLYZOS BOOSALIS/	Date Considered	08/25/2021

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

[1] See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. [2] Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). [3] For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. [4] Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. [5] Applicant is to place a check mark here if English language translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /F.P.B./

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Attorney Docket Number	4016-1056

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Yun H. Choe/	Date (YYYY-MM-DD)	2020-12-04
Name/Print	Yun H. Choe	Registration Number	61798

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

/FANI POLYZOS BOOSALIS/ 08/25/2021

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Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these record s.

2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.

3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.

4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).

5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.

6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).

7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.

8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.

9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

/FANI POLYZOS BOOSALIS/ 08/25/2021

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /F.P.B./

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (01-10)
Approved for use through 07/31/2012. OMB 0651-0031
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE
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	1	101730887	KR	B1	2017-04-28	FTLAB CO., LTD.	English Abstract	
	2	101730891	KR	B1	2017-04-28	FTLAB CO., LTD.	English Abstract	
	3	10-1935880	KR	B1	2019-01-17	ISQUARE, INC.	English Abstract	

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT

(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	KO, Jae Jun
Art Unit	TBA
Examiner Name	TBA
Attorney Docket Number	4016-1056

	4	10-2015-0093987	KR	A	2015-08-19	LISOTEK CO., LTD. et al.	English Abstract	
	5	10-2017-0023599	KR	A	2017-03-06	FTLAB CO., LTD.	English Abstract	
	6	2015-156463	JP	A	2005-06-16	MITSUBISHI HEAVY IND., LTD.	English Abstract	
	7	10-2018-0129480	KR	A	2018-12-05	LEE, Hoon	English Abstract	

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T[5]
	1	International Search Report and Written Opinion for International application No. PCT/KR2020/000837, April 24, 2020, ISA/KR	

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[1] See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. [2] Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). [3] For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. [4] Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. [5] Applicant is to place a check mark here if English language translation is attached.

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CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Yun H. Choe/	Date (YYYY-MM-DD)	2020-12-04
Name/Print	Yun H. Choe	Registration Number	61798

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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